

Cycle vs event rebuild time for three planners

A typical rebuild is far below the 100 ms limit (95 % of rebuilds finish within 13–43 ms). Rebuilds over 100 ms are rare (at most 77 per 10,000). Each rebuild is a new chance of one, so rebuilding less often stops fewer robots: at 20 ms checks, cycle vs event Voronoi 4 vs 3, PRM 25 vs 6, Visibility Graph 55 vs 22. Rebuild time is mostly map construction; the Dijkstra search is a few percent. Times are wall clock on one machine, successful rebuilds only. ▲ blue (bold) = better of the pair, ▼ orange = worse; times marked only when they differ by 10 % or more.

	Rebuilds per run	Time per rebuild, median (ms)	95 % of rebuilds under (ms)	99 % of rebuilds under (ms)	Slowest rebuild (ms)	Rebuilds over 100 ms per 10,000	Share of time in Dijkstra search (%)	Robots stopped by 100 ms limit
Voronoi								
Cycle @ 20 ms	110.4 ▼	5.6	13.2 ▲	22.2 ▲	532 ▼	2 ▲	1	4 ▼
Event (ours) @ 20 ms	9.1 ▲	5.7	17.1 ▼	35.4 ▼	250 ▲	16 ▼	1	3 ▲
Cycle @ 100 ms	22.0 ▼	5.8	13.8	30.5	215 ▼	11 ▼	1	5 ▼
Event (ours) @ 100 ms	5.2 ▲	5.8	14.1	30.6	98 ▲	0 ▲	1	0 ▲
PRM								
Cycle @ 20 ms	75.9 ▼	10.6	25.1 ▲	51.1 ▲	340 ▲	16 ▲	1	25 ▼
Event (ours) @ 20 ms	7.7 ▲	10.6	30.8 ▼	56.3 ▼	456 ▼	39 ▼	1	6 ▲
Cycle @ 100 ms	14.9 ▼	10.5	21.8	55.9 ▲	219 ▲	44 ▲	1	13 ▼
Event (ours) @ 100 ms	4.6 ▲	10.6	21.8	77.0 ▼	387 ▼	77 ▼	1	7 ▲
Visibility Graph								
Cycle @ 20 ms	107.6 ▼	13.6	28.3 ▲	58.0 ▲	702 ▼	26 ▲	2	55 ▼
Event (ours) @ 20 ms	23.2 ▲	14.2	43.2 ▼	69.1 ▼	424 ▲	47 ▼	3	22 ▲
Cycle @ 100 ms	21.0 ▼	13.9	35.0 ▲	57.5	818 ▼	29 ▼	2	12 ▼
Event (ours) @ 100 ms	8.3 ▲	13.7	39.4 ▼	60.4	227 ▲	24 ▲	2	4 ▲